Bahria University, Islamabad

Department of Software Engineering

Artificial Intelligence Lab (Spring-2026)

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Enrollment : 01-131232-089

Lab Journal: 01

Date: 30th January, 2026

Task No:	Task Wise Marks		Documentation Marks		Total Marks (20)
	Assigned	Obtained	Assigned	Obtained	
1	3		5		
2	3				
3	3				
4	3				
5	3				

Comments:

Signature

Lab No: 1 – Getting Started with Python

Tools Used

- **Python3**
- **VS-Code**

Task 1:

Code

```python
# Program to evaluate grade according to marks

def evaluate_grade(marks):
    """
    Evaluates and returns the grade based on marks.
    """
    if marks < 0 or marks > 100:
        return "Invalid marks! Please enter marks between 0 and 100."
    elif marks >= 90:
        return "A+"
    elif marks >= 80:
        return "A"
    elif marks >= 70:
        return "B"
    elif marks >= 60:
        return "C"
    elif marks >= 50:
        return "D"
    else:
        return "F"


def main():
    try:
        print("\n-------- Grade Evaluation Program -------\n")
        marks = float(input("Enter your marks (0-100): "))
        grade = evaluate_grade(marks)
        print(f"Your grade is: {grade}\n")
    except ValueError:
        print("\nInvalid input! Please enter a numeric value.")


if __name__ == "__main__":
    main()
```

Screenshot

```
● Talhas-MacBook-Air:Lab 01 talha$ python3 Task.py

  -------- Grade Evaluation Program -------

  Enter your marks (0-100): 79
  Your grade is: B
```

Figure 1. Output

```
● Talhas-MacBook-Air:Lab 01 talha$ python3 Task.py

  -------- Grade Evaluation Program -------

  Enter your marks (0-100): 84
  Your grade is: A
```

Figure 2. Output

```
● Talhas-MacBook-Air:Lab 01 talha$ python3 Task.py

  -------- Grade Evaluation Program -------

  Enter your marks (0-100): 45
  Your grade is: F
```

Figure 3. Output

Conclusion

In this lab, I successfully installed Python and familiarized myself with its core features as a high-level, interpreted language. I explored the concept of variables, specifically noting how Python handles dynamic typing without the need for explicit declarations, and I practiced casting data to specific types when necessary. I also learned to manipulate strings using various quoting methods and escape characters to ensure correct output formatting. Furthermore, I implemented flow control using `if`, `elif`, and `else` statements to direct the logic of my programs. Finally, I applied these fundamental concepts to solve practical exercises, such as creating a program to evaluate grades